Prateek Prakash Bolagond

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Professional Summary

Electronics and Communication Engineering student with an interest in Analog IC Design and CMOS circuit design. Familiar with designing and simulating analog circuits using Cadence Virtuoso and Spectre. Strong foundation in semiconductor devices, MOSFET operation, and analog circuit fundamentals. Seeking an Layout Design Internship to apply my knowledge and gain practical industry experience.

Education

University of Visvesvaraya College of Engineering (UVCE) **2023–2027**
Bachelor of Technology in Electronics and Communication Engineering [CGPA: 8.18/10] Bengaluru, Karnataka

Tungal Science Composite P U College **2021–2023**
Pre-University Course (Class XII) [95.0%] Jamkhandi, Karnataka

DPK Central School (CBSE) **2018–2021**
10th Grade [89.02%] Bagewadi, Karnataka

Technical Skills

EDA Tools: Cadence Virtuoso, LTspice, MATLAB.

Core Knowledge: CMOS VLSI Design, CMOS Device Physics, Semiconductor Devices, Electronic Circuit Design, Analog CMOS Design , Small-Signal Analysis.

Analog Circuit Fundamentals: Current Mirrors, Differential Amplifiers, Operational Amplifiers (OTA/Op-Amp), Bias Circuits, Bandgap Reference (BGR) Fundamentals, LDO Fundamentals, ADC/DAC Fundamentals.

Programming Languages: C, Python, Verilog HDL.

Academic Projects

5T Operational Transconductance Amplifier (OTA) Design Cadence Virtuoso, Spectre, GPDK45

- Designed a 5-transistor CMOS OTA in GPDK45 technology, implementing a differential input stage with a current mirror active load.
- Performed comprehensive DC, AC, Transient, and Operating Point analyses using Spectre to evaluate gain, phase margin, slew rate, and output swing.
- Optimized transistor device dimensions and sizing ratios to maximize key analog performance parameters while successfully reducing overall power consumption.

CMOS Bandgap Reference Circuit Design Cadence Virtuoso, Spectre, GPDK45

- Designed a CMOS Bandgap Reference circuit utilizing PTAT (Proportional to Absolute Temperature) and CTAT (Complementary to Absolute Temperature) voltage generation techniques.
- Implemented current mirrors and resistor networks to achieve a stable, temperature-independent reference voltage.
- Executed DC and Temperature Sweep simulations ($-30°C$ to $150°C$) to validate voltage stability and verify startup operation for reliable performance.
- Optimized resistor ratios to ensure circuit robustness across extreme thermal variations.

Certifications

- Automateum 2.0 – Hands-on IoT Workshop – BMSCE IEEE PES & Sensors Council, November 2024.
- Basics of VLSI Design – Campus Champs, 2025.